



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

January 23, 2026

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Fluid Elements

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial \mathbf{t}$ for any measurable quantity Q

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial t$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via D/Dt as the rate of change

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial t$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via D/Dt as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial\mathbf{t}$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via $D/D\mathbf{t}$ as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

Continuous Variables

Continuous Variables

- *Primitive Fields:*

Continuous Variables

- *Primitive Fields:*
 - velocity u (vector)

Continuous Variables

- *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)

- pressure p (scalar)

Continuous Variables

- *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)

- density ρ (scalar)

- pressure p (scalar)

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

Continuous Variables

- *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)

- pressure p (scalar)
- temperature T (scalar)

- *Conserved Quantities:*

Continuous Variables

- *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)

- pressure p (scalar)
- temperature T (scalar)

- *Conserved Quantities:*

- mass M (scalar)

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)

- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)

- momentum $\mathbf{P}$ (vector)

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity η (scalar)

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity η (scalar)
- vorticity $\boldsymbol{\omega}$ (vector)

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Dimensionless Numbers

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)
 - ▶ $40 \lesssim R \lesssim 4000$: Transitional Kármán Street (vortex shedding)

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)
 - ▶ $40 \lesssim R \lesssim 4000$: Transitional Kármán Street (vortex shedding)
 - ▶ $R > 4000$: Kolmogorov Energy Cascade (fully developed phase)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
- $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
- $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)
- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

Bulk Transport

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

$$\frac{\partial u}{\partial t} + (\mathbf{v} \cdot \nabla) u = 0$$

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} = 0$$

- *Nonlinear Advection*: velocity field $\mathbf{u}$ is carried by itself instead of fixed background flow

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} = 0$$

- *Nonlinear Advection*: velocity field $\mathbf{u}$ is carried by itself instead of fixed background flow

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = 0$$

Linear Advection

Nonlinear Advection

Stochastic Transport

Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation*: combines shock generation with counteracting diffusive term

Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation*: combines shock generation with counteracting diffusive term

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = \nu \nabla^2 u$$

Diffusion

Bateman-Burgers Equation

Equilibrium Search

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- *Gravity*: $\nabla^2 \Phi = 4\pi G \rho$

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- *Gravity*: $\nabla^2 \Phi = 4\pi G \rho$
- *Electrostatics*: $\nabla^2 V = -\rho / \epsilon$

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- Gravity: $\nabla^2 \Phi = 4\pi G \rho$
- Electrostatics: $\nabla^2 V = -\rho / \epsilon$
- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla) u)$

Laplace Equation

Poisson Equation

Navier-Stokes Equations

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Lid Driven Cavity Flow

Pressure Driven Channel Flow

Nonuniform Periodic Flow

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries
 - strong numerical diffusion at low orders

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries
 - strong numerical diffusion at low orders
 - inflexible rectilinear grid with fixed resolution

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries
 - strong numerical diffusion at low orders
 - inflexible rectilinear grid with fixed resolution
 - mass and energy can leak due to numerical errors

Integration Techniques

Integration Techniques

- *Spatial:*

Integration Techniques

- *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation

Integration Techniques

- *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
 - *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive

Integration Techniques

- *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

■ *Temporal:*

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

■ *Temporal:*

- *Backward Euler:* looks to future to calculate next step implicitly by inversion, extremely stable

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

■ *Temporal:*

- *Backward Euler:* looks to future to calculate next step implicitly by inversion, extremely stable
- *Forward Euler:* looks to past to find next step explicitly, simple but unstable for large intervals

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

■ *Temporal:*

- *Backward Euler:* looks to future to calculate next step implicitly by inversion, extremely stable
- *Forward Euler:* looks to past to find next step explicitly, simple but unstable for large intervals
- *Leapfrog:* reaches back to previous step instead of current one, symplectic and invertible

Integration Techniques

- *Precision Methods:*

Integration Techniques

- *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order

Integration Techniques

- *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump

Integration Techniques

■ *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information

Integration Techniques

■ *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information
- *Leapfrog*: preserve phase-space volume and total energy, enable backward reconstruction

Integration Techniques

■ *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information
- *Leapfrog*: preserve phase-space volume and total energy, enable backward reconstruction

■ *Courant-Friedrichs-Lewy Condition*: $\frac{v\Delta t}{\Delta x} \leq 1$

Integration Techniques

■ *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information
- *Leapfrog*: preserve phase-space volume and total energy, enable backward reconstruction

■ *Courant-Friedrichs-Lewy Condition*: $\frac{v\Delta t}{\Delta x} \leq 1$

Integration Techniques

■ *Precision Methods:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information
- *Leapfrog*: preserve phase-space volume and total energy, enable backward reconstruction

■ *Courant-Friedrichs-Lewy Condition*: $\frac{v\Delta t}{\Delta x} \leq 1$

■ *Pure Diffusion Condition*: $\frac{2v\Delta t}{\Delta x^2} \leq 1$

Discretized Formulation

Discretized Formulation

■ *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$

Discretized Formulation

- *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$
- *Diffusion:* $\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

Discretized Formulation

■ *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$

■ *Diffusion:* $\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

■ *Bateman-Burgers Equation:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

Discretized Formulation

■ *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$

■ *Diffusion:* $\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

■ *Bateman-Burgers Equation:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

■ *Laplace Equation:* $\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \longrightarrow \frac{p_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} = 0$

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well
 - easily deals with shock discontinuities

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well
 - easily deals with shock discontinuities
- Drawbacks:

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well
 - easily deals with shock discontinuities
- Drawbacks:
 - mathematically complex at higher order accuracy

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries
 - handles extremely complex mesh geometries well

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries
 - handles extremely complex mesh geometries well
- Drawbacks:

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries
 - handles extremely complex mesh geometries well
- Drawbacks:
 - computationally expensive and memory intensive

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries
 - handles extremely complex mesh geometries well
- Drawbacks:
 - computationally expensive and memory intensive
 - bad stability for fast turbulent flows

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Outlook

Outlook

- *Frameworks:*

Outlook

- *Frameworks:*
 - Smoothed Particle Hydrodynamics (SPH)

Outlook

- *Frameworks:*

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)

Outlook

■ *Frameworks:*

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

Outlook

- *Frameworks:*

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

- *Regimes:*

Outlook

- *Frameworks:*

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

- *Regimes:*

- *Plasma:* Magnetohydrodynamics (MHD)

Outlook

- *Frameworks:*

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

- *Regimes:*

- *Plasma:* Magnetohydrodynamics (MHD)
- *Relativity:* Relativistic Hydrodynamics (RHD)

Outlook

■ *Frameworks:*

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ *Regimes:*

- *Plasma:* Magnetohydrodynamics (MHD)
- *Degeneracy:* Quantum Effects
- *Relativity:* Relativistic Hydrodynamics (RHD)

Outlook

■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- | | |
|--|--|
| ■ <i>Plasma</i> : Magnetohydrodynamics (MHD) | ■ <i>Degeneracy</i> : Quantum Effects |
| ■ <i>Relativity</i> : Relativistic Hydrodynamics (RHD) | ■ <i>Collisionless</i> : Initial Distributions |

Outlook

■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- | | |
|--|--|
| ■ <i>Plasma</i> : Magnetohydrodynamics (MHD) | ■ <i>Degeneracy</i> : Quantum Effects |
| ■ <i>Relativity</i> : Relativistic Hydrodynamics (RHD) | ■ <i>Collisionless</i> : Initial Distributions |

■ Applications:

Outlook

■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- *Plasma*: Magnetohydrodynamics (MHD)
- *Relativity*: Relativistic Hydrodynamics (RHD)
- *Degeneracy*: Quantum Effects
- *Collisionless*: Initial Distributions

■ Applications:

- Stellar Evolution

Outlook

■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- *Plasma*: Magnetohydrodynamics (MHD)
- *Relativity*: Relativistic Hydrodynamics (RHD)
- *Degeneracy*: Quantum Effects
- *Collisionless*: Initial Distributions

■ Applications:

- Stellar Evolution
- Accretion Disks

Outlook

■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- *Plasma*: Magnetohydrodynamics (MHD)
- *Relativity*: Relativistic Hydrodynamics (RHD)
- *Degeneracy*: Quantum Effects
- *Collisionless*: Initial Distributions

■ Applications:

- Stellar Evolution
- Accretion Disks
- Cosmic Filaments

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)